

SQL Aggregate Functions and Operators

Question 1

```
SELECT DISTINCT department  
FROM students;
```

~~IT~~

HR

FINANCE

department
IT
HR
FINANCE

Question 2

```
SELECT department, AVG(age) AS avg_age  
FROM students  
GROUP BY department
```

department	avg_age
IT	20.5
HR	22
FINANCE	23

Question 3

```
SELECT department, COUNT(*) AS student_count  
FROM students  
GROUP BY department  
HAVING COUNT(*) > 1
```

department	student count
IT	2
HR	2

Question 4

```
SELECT student_id, name, age, department
FROM students
WHERE age BETWEEN 21 AND 23;
```

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	FINANCE
5	Eve	22	HR

Question 5

```
SELECT student_id, name, age, department
FROM students
WHERE department IN ('IT', 'HR')
AND age > 21;
```

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

Question 6

```
SELECT department, SUM(credits) AS total_credits
FROM courses
GROUP BY department
HAVING sum (credits) > 5;
```

department	total credits
IT	11

Question 7

```
SELECT course_id, course_name, department, credits
FROM courses
WHERE credits < > 4
```

course_id	course_name	department	credits
101	SQL BASICS	IT	3
104	EXCEL	Finance	2
105	Statistics	HR	3

Question 8

```
SELECT course_id, course name, credits
FROM courses
ORDER BY credits DESC
LIMIT 3;
```

course_id	course name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

Question 9

```
SELECT  
MAX (grade) AS max_grade,  
MIN (grade) AS min_grade,  
AVG (grade) AS avg_grade  
FROM enrollments;
```

max_grade	min_grade	avg_grade
90	78	84.6

Question 12

```
SELECT department, AVG(salary) AS avg_salary  
FROM salaries  
GROUP BY department  
HAVING AVG(salary) > 55000;
```

department	avg_salary
IT	61000
Finance	70000

Question 13

```
SELECT employee_id, name, salary, bonus, (salary + bonus)  
AS total_compensation  
FROM salaries  
WHERE (salary + bonus) > 60000;
```

employee_id	name	salary	bonus	total_compensation
1	TOM	60000	5000	65000
3	SPIKE	70000	6000	76000
4	Tyke	62000	5500	67500

• Question 14

```

SELECT department
SUM (budget) AS total_budget,
AVG (budget) AS avg_budget
FROM projects
GROUP BY department
HAVING AVG (budget) > 70000

```

department	total_budget	avg_budget
IT	270000	135000
Finance	80000	80000

Question 15

```

SELECT project_id, project_name, department,
budget
FROM projects
WHERE budget BETWEEN 50000 AND 120000
AND department <> 'marketing';

```

project_id	project_name	department	budget
1	AI App	IT	120000
2	Payroll System	Finance	80000
5	HR Portal	HR	50000